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Optimal filter bandwidth for pulse oximetry

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Pulse oximeters contain one or more signal filtering stages between the photodiode and microcontroller. These filters are responsible for removing the noise while retaining the useful frequency components of the signal, thus improving the signal-to-noise ratio. The corner frequencies of these filters affect not only the noise level, but also the shape of the pulse signal. Narrow filter bandwidth effectively suppresses the noise; however, at the same time, it distorts the useful signal components by decreasing the harmonic content. In this paper, we investigated the influence of the filter bandwidth on the accuracy of pulse oximeters. We used a pulse oximeter tester device to produce stable, repetitive pulse waves with digitally adjustable R ratio and heart rate. We built a pulse oximeter and attached it to the tester device. The pulse oximeter digitized the current of its photodiode directly, without any analog signal conditioning. We varied the corner frequency of the low-pass filter in the pulse oximeter in the range of 0.66–15 Hz by software. For the tester device, the R ratio was set to $R = 1.00$, and the R ratio deviation measured by the pulse oximeter was monitored as a function of the corner frequency of the low-pass filter. The results revealed that lowering the corner frequency of the low-pass filter did not decrease the accuracy of the oxygen level measurements. The lowest possible value of the corner frequency of the low-pass filter is the fundamental frequency of the pulse signal. We concluded that the harmonics of the pulse signal do not contribute to the accuracy of pulse oximetry. The results achieved by the pulse oximeter tester were verified by human experiments, performed on five healthy subjects. The results of the human measurements confirmed that filtering out the harmonics of the pulse signal does not degrade the accuracy of pulse oximetry. © 2012 American Institute of Physics. [<http://dx.doi.org/10.1063/1.4759491>]

I. INTRODUCTION

A photoplethysmographic waveform can be reproduced using the first ten harmonics of a pulse signal. The heart rate is determined by the fundamental frequency of the pulse signal. In addition to the heart rate, photoplethysmographic waveforms contain valuable information used by professionals. Specifically, this waveform can be used to measure the degree of one's health. For instance, it was observed that the subject's age is correlated with the third and seventh harmonics.¹ Further, photoplethysmographic waveforms can be used to assess venous insufficiency.² The filters of signal processing circuits must suppress the noise in photoplethysmograph waveforms, while preserving the useful frequency region. Because the harmonics of the photoplethysmographic waveform are of crucial importance and should be retained, it is recommended to use a filter bandwidth of 0.01–15 Hz.³ By using two photoplethysmographic waveforms with different wavelengths, the oxygen saturation of arterial blood can be determined on the basis of the principles of pulse oximetry.⁴ It is a well known issue in pulse oximeters that the movement of a patient during the measurement process creates motion artifacts in the signal.⁵ Because the frequency components of motion artifact overlap those of the pulse oximeter signal, it is difficult to filter out this artifact.^{6,7} Several attempts were made to solve the problem of motion artifact.^{8–11} Reducing the bandwidth of the pulse signal used for data evaluation reduces the chance of overlapping. Furthermore, narrowing the bandwidth of the

filter of the photodiode signal has a positive effect on noise suppression. However, reducing the signal bandwidth may affect the accuracy of oxygen level measurement. In this paper, we investigated the influence of the signal bandwidth on the accuracy of a pulse oximeter.

In Figure 1, we present a simplified schematic of a pulse oximeter. The light is generated by (at least) two light-emitting diodes (LEDs) with different wavelengths, typically in the region of 650–950 nm. The light beam passing through living tissues is modulated by arterial pulsation and is finally sensed by a photodiode. In Figure 2(a), we present an example waveform of a sensed, unfiltered signal. The pulse oximeter measures the ratio of the amplitudes of two such pulse signals. The rate of the amplitudes is called the R ratio and is expressed as¹²

$$R = \frac{\ln \left(\frac{I_{\max}(\lambda_1)}{I_{\min}(\lambda_1)} \right)}{\ln \left(\frac{I_{\max}(\lambda_2)}{I_{\min}(\lambda_2)} \right)}, \quad (1)$$

where λ_1 and λ_2 are the two (different) measurement wavelengths (e.g., 650 nm and 910 nm), and $I_{\max}$ and $I_{\min}$ are the maximum and minimum intensities of a pulse, respectively (i.e., the maximum and minimum points of the curve in Figure 2). By using (1), pulse oximeters measure the R ratio of each pulse and average 4–32 consecutive values to decrease measurement uncertainty. Next, the arterial-blood-oxygen saturation is calculated from the averaged R ratio and displayed to the user.

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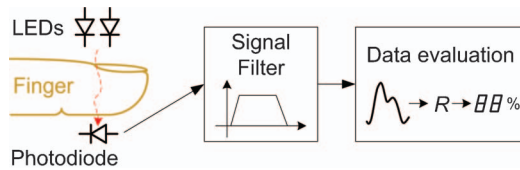


FIG. 1. Simplified schematic of a pulse oximeter. Measuring head is attached to the skin surface (left side). LEDs generate light, which passes through the tissues, and a photodiode senses the light. After signal filtering and digitizing, an algorithm implemented on a digital signal processor (DSP) or PC calculates the blood-oxygen level from the measured R ratio.

The signal path of pulse oximeters contains one or more filters, which are used to filter out the noise and unwanted signal components from the received signal (Figure 1). These filters can be analog, digital, or both. The narrower the bandwidth of the filter, the lower the noise content of the filtered signal. Noise comes from the internal electronic components of the pulse oximeter and from external sources. The photodiode of the pulse oximeter also senses environmental light, which flickers in the case of fluorescent lighting. Implementing a signal filter with a narrow bandwidth appears to be an obvious solution for separating the pulse signal from other signal components, such as noise (Figure 2). However, by lowering the corner frequency of a low-pass filter (LPF) in the signal path, we also decrease the information content of the useful signal. As shown in Figure 2(d), the narrow bandwidth of the filter causes the dicrotic notch to disappear, and the filtered signal resembles a sinusoidal wave than a pulse wave. Cutting off the harmonics of the measured pulse signal significantly distorts the shape of the signal and may affect the R ratio, and consequently, the oxygen saturation measurement. In this paper, we determine the lowest possible LPF corner frequency that preserves the accuracy of the oxygen level measurement.

II. METHODS

A measurement setup was arranged to investigate the influence of the filter bandwidth on the accuracy of the arterial-blood-oxygen level measurement. A schematic of the measurement setup is presented in Figure 3. The pulse oximeter tester device (SpO₂ 5002, Medihead), on the left side, was used to continuously produce artificial pulse waves. This device stored a standard pulse waveform in a non-volatile memory. By controlling the light intensity of the LED on the measuring head of the tester device, pulse waves were optically transmitted to the attached oximeter. The tester device synchronized its light transmission by sensing the LED activity

of the attached oximeter. The measuring head of the tester device was equipped with a photodiode that was sensitive to visible light (“PD” in Figure 3) and a photodiode that was sensitive to infrared light (“IR PD” in Figure 3). Therefore, the tester device was capable of differentiating between the two measuring wavelengths of the attached pulse oximeter. The tester device was generating pulsatile waveforms as if it were real living tissue attached to the oximeter. The R ratio and heart rate were digitally adjustable on the tester device. The attached oximeter used in the experiments was a direct current converting pulse oximeter, which was developed by the authors. No analog components were present between the photodiode and analog-to-digital converter (ADC) in the pulse oximeter. Because the current of the photodiode was directly digitized without any analog signal conditioning, it was possible to observe the real photodiode signal. Filtering and signal modification was performed only in the digital domain. Because all parameters of the signal conditioning chain of the pulse oximeter were adjustable by software, we were able to evaluate the influence of the filter parameters on the pulse oximeter signal in a well controlled and precise manner. After the personal computer (PC) received the measured data through a radio link (right side in Figure 3), the signal chain was divided into two channels: red and infrared. To determine the arterial-blood-oxygen level, we must first calculate the R ratio. Therefore, we applied the natural logarithm function to both channels according to (1). Next, the two channels were filtered using adjustable digital filters denoted by red text in Figure 3. After filtering, an algorithm was applied to determine the maximum and minimum values of each pulse. The difference between these values represented the pulse amplitude. The rate of the amplitudes is equal to the R ratio, and was computed according to (1). The evaluation software was created in MATLAB Simulink.

The adjustable digital filter consisted of a high-pass filter (HPF) and a LPF. The HPF was a second-order infinite impulse response (IIR) filter with a corner frequency of 0.1 Hz. The purpose of the HPF was to remove the DC component of the signal. The corner frequency of the HPF must be lower than the fundamental frequency of the pulse. The LPF was a 100th-order finite impulse response (FIR) filter and its corner frequency was adjusted between 0.66 Hz and 15 Hz in five steps (0.66 Hz, 1 Hz, 1.5 Hz, 3 Hz, and 15 Hz). The stop-band attenuation of the filters was set to 96 dB. The sample frequency was 40 Hz on both channels.

During measurements, the R ratio on the tester device was set to $R = 1.00$. Moreover, the tester device generated pulse waves with fundamental frequencies of 0.66, 1, and

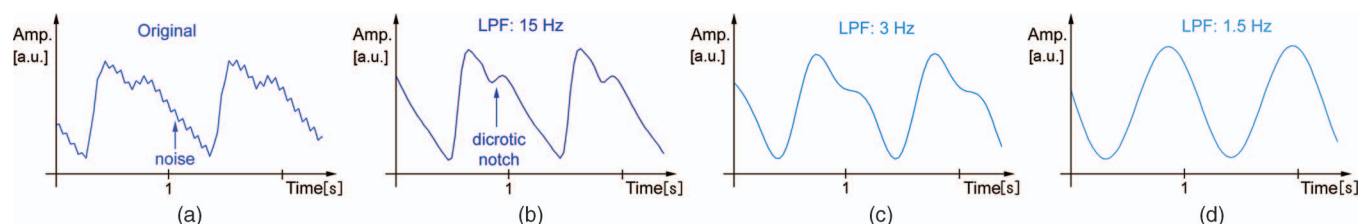


FIG. 2. (a) Pulse waveform sensed on a forefinger at 870 nm, using a reflective type pulse oximeter; heart rate of the subject was 0.83 Hz (72 bpm); (b)–(d) original signal filtered by a LPF with corner frequencies 15 Hz, 3 Hz, and 1.5 Hz, respectively.

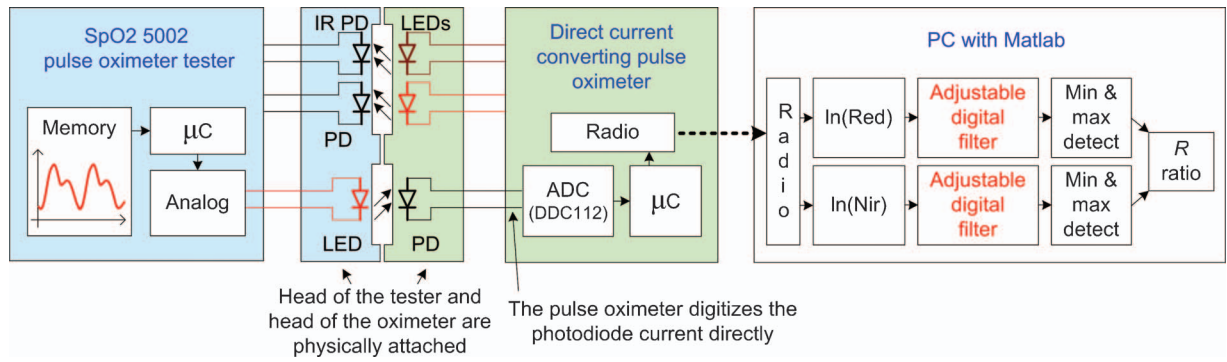


FIG. 3. Measurement setup used to investigate the influence of the filter bandwidth on the accuracy of the arterial-blood-oxygen level reading. The setup consists of a pulse oximeter tester (left side, blue background), which optically generates artificial pulse waves with a predefined R ratio and heart rate. A pulse oximeter was used to sense a signal that was an imitation of a signal from real-living tissue (middle part, green background). The measured data were evaluated by an algorithm implemented in MATLAB Simulink (right side).

1.5 Hz (corresponding to a heart rate of 40, 60, and 90 bpm, respectively). At each filter frequency and heart rate, we recorded and evaluated 32 pulses. The deviation of the R ratio (difference from 1.00) was recorded for all corner frequencies (0.66 Hz, 1 Hz, 1.5 Hz, 3 Hz, and 15 Hz) and heart rates (0.66 Hz, 1 Hz, and 1.5 Hz). The deviation from the $R = 1.00$ value indicates the error in the measured oxygen level (caused by lowering the LPF corner frequency). By observing the R ratio deviations at different filter frequencies, we determined the influence of the filter bandwidth on the accuracy of the oxygen level measurements. In addition to the R ratio, we also recorded the amplitude of the filtered pulse signal.

To verify the results obtained with the oximeter tester, we performed R ratio measurements on humans. Specifically, we measured the R ratio of five healthy adults (ages: 27 ± 4 years) for 75 s, who were breathing normal atmospheric air. The same direct current converting pulse oximeter was attached to the forefinger of each subject. Pulse oximeters calculate the SpO_2 values from the measured R ratios by using a

look-up table or calibration curve.¹³ By applying our noninvasive pulse oximeter calibration method published earlier,¹⁴ we determined the SpO_2 levels corresponding to the measured R ratios. The corner frequency of the adjustable filters was varied using the same steps as mentioned previously (15 Hz, 3 Hz, 1.5 Hz, and 1 Hz). The SpO_2 value measured with a LPF corner frequency of 15 Hz was chosen to be the reference value, and deviations of the SpO_2 values from the reference value were recorded for the corner frequencies of 1 Hz, 1.5 Hz, and 3 Hz. Because the heart rate of all subjects was higher than 0.66 Hz (40 bpm), the 0.66-Hz filter frequency could not be used during human measurements.

III. RESULTS

In Figure 4, we present the results of the R ratio measurements obtained by the pulse oximeter tester device. We divide the measured R ratios (green dots) into five groups according to the corner frequency of the LPF ($f_{\text{LPF}} = 15$ Hz, 3 Hz, 1.5 Hz, 1 Hz, and 0.66 Hz). Each group

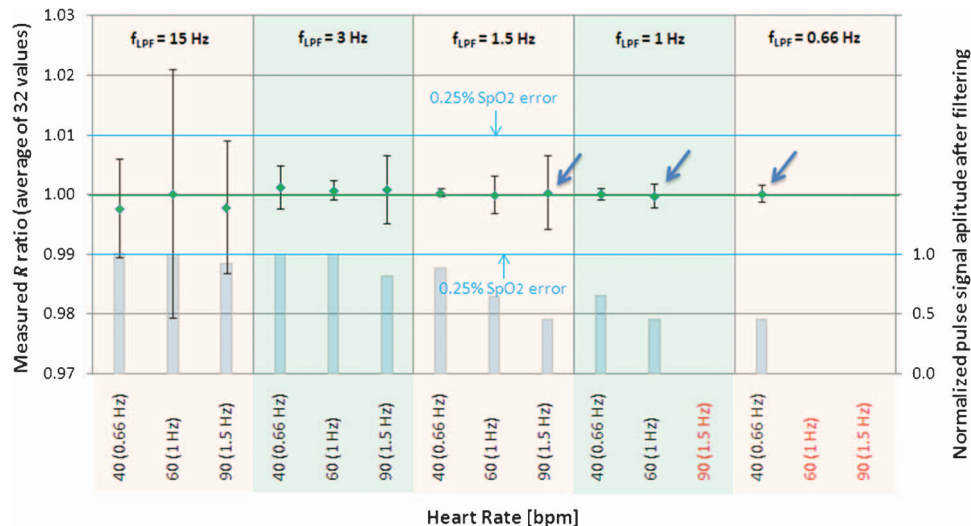


FIG. 4. Measured R ratios as a function of the LPF corner frequency (f_{LPF}) applied to the pulse signal and as a function of the heart rate (x-axis). Each measurement point is the average of 32 R ratio values.

TABLE I. Results of the oxygen level measurements performed on human subjects. The photodiode signal was filtered with a LPF. The corner frequency was varied in four steps (15 Hz, 3 Hz, 1.5 Hz, and 1 Hz). The SpO₂ value measured with a corner frequency of 15 Hz was used as a reference value for each subject. The SpO₂ deviations caused by narrowing the LPF were recorded.

Subject	HR	Reference% SpO ₂ (LPF corner freq = 15 Hz)		Deviation at LPF = 3 Hz [% SpO ₂]	Deviation at LPF = 1.5 Hz [% SpO ₂]	Deviation at LPF = 1 Hz [% SpO ₂]
1	1.3 Hz (77 bpm)	98.27 ± 2.46	($R = 0.645 \pm 0.035$)	−0.35	−0.28	HR is higher
2	1.2 Hz (73 bpm)	97.72 ± 0.98	($R = 0.695 \pm 0.039$)	−0.41	−0.46	HR is higher
3	1.1 Hz (66 bpm)	99.29 ± 0.74	($R = 0.632 \pm 0.03$)	−0.17	−0.44	HR is higher
4	1 Hz (60 bpm)	98.73 ± 0.88	($R = 0.673 \pm 0.054$)	0.02	−0.06	0.23
5	1.4 Hz (81 bpm)	97.06 ± 1.02	($R = 0.721 \pm 0.041$)	−0.30	−0.32	HR is higher
<i>Average</i>				−0.24	−0.31	0.23

contains three measured R ratio values, corresponding to heart rates of 1.5 Hz, 1 Hz, and 0.66 Hz (90 bpm, 60 bpm, and 40 bpm, respectively). Each dot represents the average of 32 consecutively measured R ratios. The standard deviation of the 32 samples was marked at each dot. If the fundamental frequency of the pulse signal was outside the filter bandwidth, the corresponding heart rate was indicated with red text, and no data were evaluated for these frequencies.

Figure 2 illustrates that the lower filter frequencies cut off the harmonics of the pulse signal. The cases where the fundamental frequency of the pulse signal was equal to the corner frequency of the filter are marked in Figure 4 with dark blue arrows. In these cases, because we filtered out all harmonics of the pulse wave, the only frequency component that contributes to the measurements is the fundamental frequency. The measured R ratio deviations for these cases were 0.0061, 0.002, and 0.0014. For all cases, the standard deviation of the 32 measured values was less than 0.01. The low values of the R ratio deviation and low standard deviations indicate that the measurement accuracy was high, even for narrow filter bandwidths; i.e., cutting off the harmonics of the pulse signal did not affect the measurement accuracy of the pulse oximeter.

In practice, the SpO₂ values are more meaningful than the R ratio values. In case of the pulse oximeter used for the measurements, a measured R ratio deviation of $\Delta R = 0.01$ corresponds to a SpO₂ error of approximately 0.25% on the display (horizontal light blue lines in Figure 4). The highest standard deviation belongs to the measurement corresponding to a 1 Hz (60 bpm) pulse wave measured by a LPF with a corner frequency of 15 Hz. In this case, the standard deviation was 0.0208, which corresponds to an error of 0.52% SpO₂. Considering that the typical measurement error in commercially available pulse oximeters is 1.5%–2% SpO₂, a 0.52% SpO₂ error is low.

For a LPF corner frequency of 15 Hz ($f_{LPF} = 15$ Hz column in Figure 4), the standard deviation at each heart rate was significantly higher. This was because the wider filter bandwidth allowed more noise to pass. Conversely, for LPF corner frequencies closer to the fundamental frequency, the standard deviations of the results were lower. The amplitudes of the filtered signal are depicted with blue bars in Figure 4. The amplitudes were normalized by the amplitude of the origi-

nal pulse signal. From Figure 4, we see that when the corner frequency of the LPF approaches the fundamental frequency, the amplitude of the filtered signal decreases, which leads to a reduced signal-to-noise ratio (SNR). This phenomenon can be mitigated by utilizing filters with steep characteristics. The results of the R ratio and SpO₂ level measurements obtained from human subjects are listed in Table I. For better understanding, we converted the measured R ratio values to SpO₂ values. The heart rate of the subjects is listed in the second column. In the third column, we list the SpO₂ levels measured with a 15-Hz corner frequency, which was chosen to be the reference SpO₂ level. In this column, we also indicate the R ratio used to calculate the reference SpO₂ level. The SpO₂ level deviations at different filter frequencies are presented in the fourth, fifth, and sixth columns. Because filter frequencies lower than the heart rate were not evaluated, the corresponding entries in the table contain the indication “HR is higher.” The deviation of the SpO₂ levels from the reference value indicates the error in the measured oxygen level (caused by narrowing the bandwidth of filtered signal). In Table I, we see that there is no observable trend in the SpO₂ deviation values. The SpO₂ deviation from the reference value was less than 0.5% SpO₂ in all cases, regardless of the LPF corner frequency. This implies that lowering the corner frequency of the LPF does not affect the measurement accuracy. The results of the human measurements confirmed the validity of the measurements performed by our tester device, i.e., filtering out the harmonics of the pulse signal does not degrade the accuracy of pulse oximetry.

IV. CONCLUSION

Lowering the corner frequency of the LPF in the pulse oximeter resulted in decreased noise, and consequently, decreased standard deviation in the measured values. Moreover, lowering the corner frequency of the LPF did not decrease the accuracy of the oxygen level measurements. The lowest possible value of the LPF corner frequency is the fundamental frequency of the pulse signal. To calculate the arterial-blood-oxygen level, it is adequate to only consider the fundamental frequency of the pulse signal. The measurement accuracy did not degrade even when all harmonics of the pulse wave were filtered out. When designing a pulse oximeter, it is advised

to set the corner frequency of the LPF as close as possible to the fundamental frequency of the pulse wave. Another advantage of decreasing the corner frequency is that the dicrotic notch disappears from the pulse signal and the signal shape becomes sinusoidal, which simplifies heart rate detection and other signal processing tasks.

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